Proteus Arc — EEG-Based Alzheimer's Classification

Dataset evaluation, literature basis and Colab pipeline documentation

Run status (read first): The Colab notebook accompanying this document was **not executed on the real ds004504 data** and **not executed in Google Colab**. The build environment could not reach OpenNeuro, and no Colab/Drive link can be created from here. The feature-extraction and cross-validation code was tested on synthetic EEG; the data-download cell is untested against the live server. No accuracy figures from this pipeline are reported in this document.

1. Dataset selection

A web search for public EEG datasets used in Alzheimer's classification identified **OpenNeuro ds004504** as the most prominent resting-state option: it is openly licensed (CC0), BIDS-formatted, ships both raw and preprocessed EEG, has clinical labels and MMSE scores, and is the basis of a data-descriptor paper, a dedicated 2026 systematic review, and multiple classification studies. Only this dataset was verified in this search; no other candidate datasets were compared.

Property	Value (from dataset description)
Title	A dataset of EEG recordings from: Alzheimer's disease, Frontotemporal dementia and Healthy subjects
ID / DOI	OpenNeuro ds004504; dataset descriptor doi:10.3390/data8060095
License	CC0
Subjects	88 total: 36 AD, 23 FTD, 29 age-matched healthy controls (CN)
Mean age (SD)	AD 66.4 (7.9); FTD 63.6 (8.2); CN 67.9 (5.4)
MMSE (mean)	AD 17.75 (SD 4.5); FTD 22.17 (SD 8.22); CN 30
Recording	Resting state, eyes closed; Nihon Kohden EEG 2100; sitting position
Channels / rate	19 scalp electrodes (10-20): Fp1, Fp2, F7, F3, Fz, F4, F8, T3, C3, Cz, C4, T4, T5, P3, Pz, P4, T6, O1, O2; 500 Hz
Duration	Approx. 13.5 min (AD), 12 min (FTD), 13.8 min (CN) per subject
Preprocessed version	Butterworth band-pass 0.5-45 Hz, re-reference to A1-A2, Artifact Subspace Reconstruction, ICA with ICLabel removal of eye/jaw components (folder derivatives/)
Format	BIDS, EEGLAB .set files; participants.tsv has Gender, Age, Group (A/F/C), MMSE

2. Evaluation of the dataset

Strengths: open license; standard BIDS layout; clinical diagnosis labels; preprocessed and raw versions; widely benchmarked, so results can be compared; ready-made participant metadata.

Weaknesses and risks:

• **Small sample (88 subjects).** Subject-level accuracy has wide uncertainty; results from a single split can mislead.
• **Class imbalance** (36 / 23 / 29), and only 23 FTD subjects.
• **Single site and single device** (one hospital), so generalisation to other clinics is untested.
• **Healthy controls have MMSE 30** while patients are impaired, which makes AD vs CN easier than real screening, where borderline cases occur.
• **No MCI group** and no longitudinal data.
• **Preprocessing (ASR/ICA) was applied by the dataset authors** to each full recording; this is common practice but should be disclosed.

• **Epoch leakage:** many windows per subject. Splitting windows randomly puts the same person in train and test and inflates accuracy. A study cited in this review found a kNN reaching 96% under 10-fold CV versus about 60% under leave-one-person-out.

3. Literature basis (papers found by web search)

The figures below are quoted from the search results (abstracts or comparison tables); they were not independently reproduced. Please verify against the original papers before citing — I do not have access to a citation database, and details may contain errors.

#	Paper	Method / validation	Reported result
1	Miltiadous et al., 'A Dataset of Scalp EEG Recordings of AD, FTD and Healthy Subjects from Routine EEG', *Data* 8(6):95, 2023. doi:10.3390/data8060095	Data descriptor	Describes dataset (no classifier benchmark)
2	Miltiadous et al., 'DICE-Net: A Novel Convolution-Transformer Architecture for Alzheimer Detection in EEG Signals', *IEEE Access* 11:71840, 2023. doi:10.1109/ACCESS.2023.3294618	Band power + coherence, 30 s windows, CNN-Transformer; Leave-One-Subject-Out	AD vs CN: 83.28% accuracy, F1 84.12%. Baselines in same paper: SVM+PCA 73.75%, kNN 72.52% (approx.), XGBoost/LightGBM ~75-76%
3	Miltiadous et al., 'AD and FTD: A Robust Classification Method of EEG Signals and a Comparison of Validation Methods', *Diagnostics* 11(8):1437, 2021. doi:10.3390/diagnostics11081437	Classical ML (DT, RF, ANN, SVM, NB, kNN); leave-one-person-out vs 10-fold. Appears to use an earlier, smaller recording set	LOPO: AD/CN decision tree 78.5%, RF 77.07%, SVM 68%; FTD/CN RF 86.3%
4	Kachare et al., 'STEADYNet: Spatiotemporal EEG analysis for dementia detection using CNN', 2024	CNN on multichannel EEG; uses two public datasets	As quoted in paper 5: 88.00% AD vs NC, 92.25% FTD vs NC
5	Khan et al., 'An explainable and efficient deep learning framework for EEG-based diagnosis of AD and FTD', *Frontiers in Medicine* 12:1590201, 2025	Modified relative band power, TCN-LSTM, SHAP	99.7% binary, 80.34% multi-class (validation protocol should be checked in the paper)
6	Mahdoubi et al., 'EEG-based ML and DL Approaches for AD and FTD: A Systematic Review of the OpenNeuro ds004504 Dataset', 2026	PRISMA review of 10 studies	Finds validation strategy has a major effect on results; notes imbalance, inconsistent evaluation, limited interpretability
7	Shamsi, 'Alzheimer's diagnosis from EEG with reliable probabilities: subject-wise, leakage-free evaluation and isotonic calibration', *J. Eng. Appl. Sci.* 72(1), 2025. doi:10.1186/s44147-025-00821-7	Listed in the review (6); only the title was seen	Not reviewed in detail
8	Conference abstract (Riemannian geometry + stacked generalisation), leave-one-subject-out on ds004504	FC features, logistic-regression meta-classifier	AD/FTD 74%, AD/HC 85%, FTD/HC 74%

Takeaway for expectations. Under strict subject-independent validation, conventional ML on ds004504 AD vs CN has been reported in roughly the high-60s to high-70s percent accuracy range, with the best deep model in paper 2 at 83%. Values above 90% usually come from different protocols or datasets. A result far above this range from the notebook should be checked for leakage before being trusted.

4. Pipeline documentation

Stage	What it does	Notebook cell
Setup	Installs libraries, defines EEG bands (delta 0.5-4, theta 4-8, alpha 8-13, beta 13-30 Hz)	1-3
Features	Per channel: mean, std, variance, RMS, min, max, range, skewness, kurtosis; Hjorth activity/mobility/complexity; Welch absolute and relative band power. 19 channels x 20 = 380 features per epoch	4, 8
Download	openneuro-py fetches participants.tsv and derivatives/ (preprocessed EEG)	5
Task choice	AD_vs_CN (default), FTD_vs_CN, AD_vs_FTD, or 3-class	6
Epoching	Reads .set with MNE, converts to microvolts, 30 s windows with 50% overlap	7
Models	Logistic Regression, RBF-SVM, Random Forest; median imputation; scaling for LR and SVM; balanced class weights	9
Validation	StratifiedGroupKFold (default) or Leave-One-Subject-Out; groups = subject ID, so no subject appears in both train and test	9
Metrics	Epoch-level accuracy, balanced accuracy, macro F1; subject-level accuracy and balanced accuracy from majority vote; sensitivity/specificity for binary tasks	9-11
Interpretation	Random Forest feature importance (fit on all data, for interpretation only)	12
Export	joblib model file	13

5. How to run in Google Colab

1. Open colab.research.google.com, choose File - Upload notebook, and select **Proteus_Arc_ds004504_EEG_Alzheimers.ipynb**.
2. Runtime - Run all (a CPU runtime is enough; the download of the derivatives folder can take several minutes).
3. If Cell 5 fails, use the AWS CLI fallback written in that cell's description.
4. To get a shareable link: File - Save a copy in Drive, then Share - Copy link.
5. Change TASK, CV_MODE (kfold or loso) and EPOCH_SEC in Cells 6 and 9 to explore other settings.

6. What was and was not tested

Component	Status
Feature extraction, epoching, task selection, CV, subject-level voting, plots, model save	Executed end-to-end on synthetic data (20 subjects, 19 channels); runs without errors. Perfect scores there only reflect the artificial signal
Dataset search and metadata	Taken from OpenNeuro/NEMAR/EEGDash search results
openneuro-py download and MNE loading of ds004504 .set files	**Not tested**; the file path pattern derivatives/sub-XXX/eeg/sub-XXX_task-eyesclosed_eeg.set follows the dataset layout reported by an independent loader but was not verified here
Run on real data / in Colab	**Not done**

7. Limitations and ethical note

Conventional ML on 19-channel resting EEG from one hospital cannot support clinical claims. Results should be reported with the validation protocol, the number of subjects per fold, and uncertainty (for example, repeated runs with different seeds). This is a research tool and not a diagnostic device.